

Why do centenarians escape or postpone cancer? The role of IGF-1, inflammation and p53.

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BACKGROUND: Centenarians are exceptionally long living individuals who escaped the most common age-related diseases. In particular they appear to be effectively protected from cancers. The mechanisms that underlie this protection are quite complex and still largely unclear. AIM: To critically analyse the literature in order to propose a unifying hypothesis that can account for this cancer protection in centenarians. METHODS: Review of the scientific literature regarding three main players in tumourigenesis such as IGF-1, inflammation and p53, and centenarians. RESULTS: Centenarians appear to be characterised by low IGF-1-mediated responses and high levels of anti-inflammatory cytokines such as IL-10 and TGF-beta, a condition that results in protection from cancer. Both inflammation and IGF-1 pathway converge on the tumour suppressor p53. Accordingly, some studies indicate that genetic variants of p53 are associated with human longevity by providing protection from cancer mortality. CONCLUSIONS: The available data let us to hypothesise that among other possible mechanisms, well-preserved p53-mediated responses are likely a key factor contributing to protection from cancer in centenarians.